

CS-860: ARTIFICIAL **INTELLIGENCE**

- **Ethics and Society**
 - **AI Everywhere**
 - **Human vs AI Characteristics**
 - **Acting Rationally**
 - **Ethics of AI**

**Text : Artificial Intelligence a Modern Approach
by Stuart Russel and Peter Norvig**

INSTRUCTOR:

DR. KUNWAR FARAZ AHMED KHAN

AI: Ethics and Society



• AI and COVID-19

AI and COVID-19



THE ROBOTREPORT

EXPLORING THE BUSINESS AND APPLICATIONS OF ROBOTICS

RESEARCH TECHNOLOGIES DEVELOPMENT ROBOTS MARKETS INVESTMENTS RESOURCES

Will COVID-19 accelerate an automated future?

By Bastiane Huang | March 29, 2020



AI: Ethics and Society



- AI and COVID-19
- **AI Everywhere**

AI Everywhere

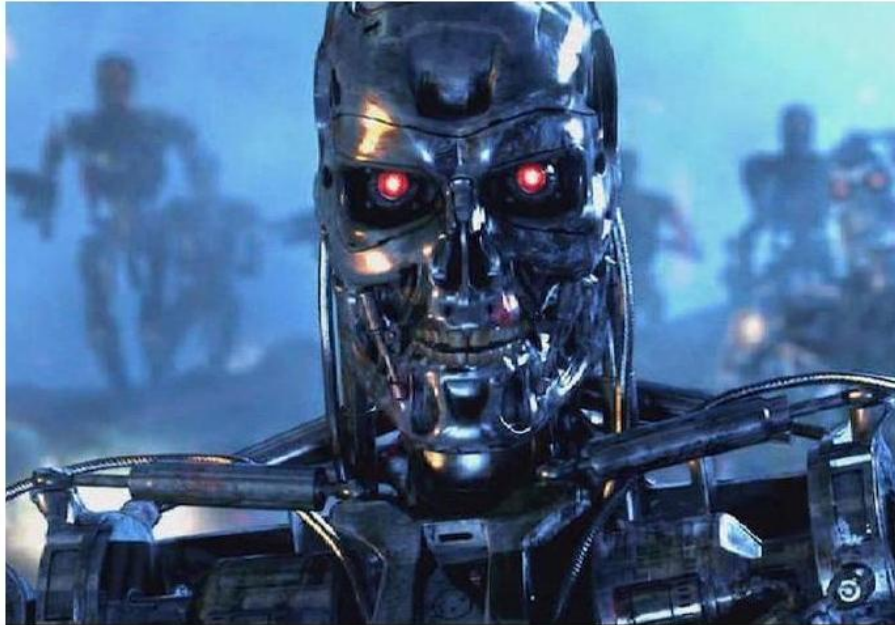


AI: Ethics and Society



- AI and COVID-19
- AI Everywhere
- **Utopia or Dystopia**

Utopia or Dystopia?



AI: Ethics and Society



- AI and COVID-19
- AI Everywhere
- **Utopia or Dystopia**

Utopia or Dystopia? – Not So Fast!



AI: Ethics and Society

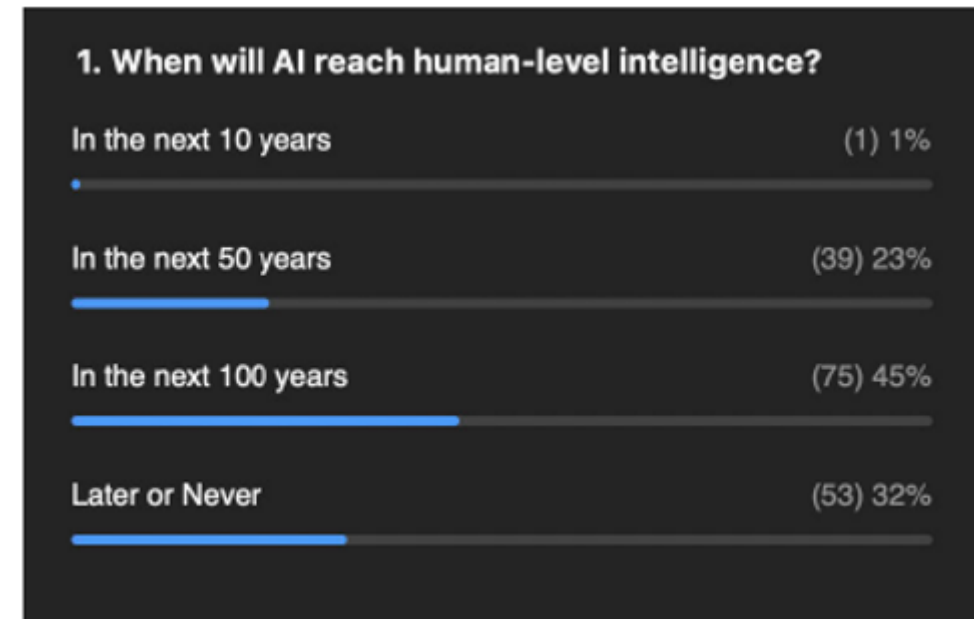


- AI and COVID-19
- AI Everywhere
- Utopia or Dystopia
- **Discussion Question**

Discussion Question

When will AI reach Human-Level Intelligence?

- In the next 10 years?
- In the next 50 years?
- In the next 100 years?
- Later or Never?



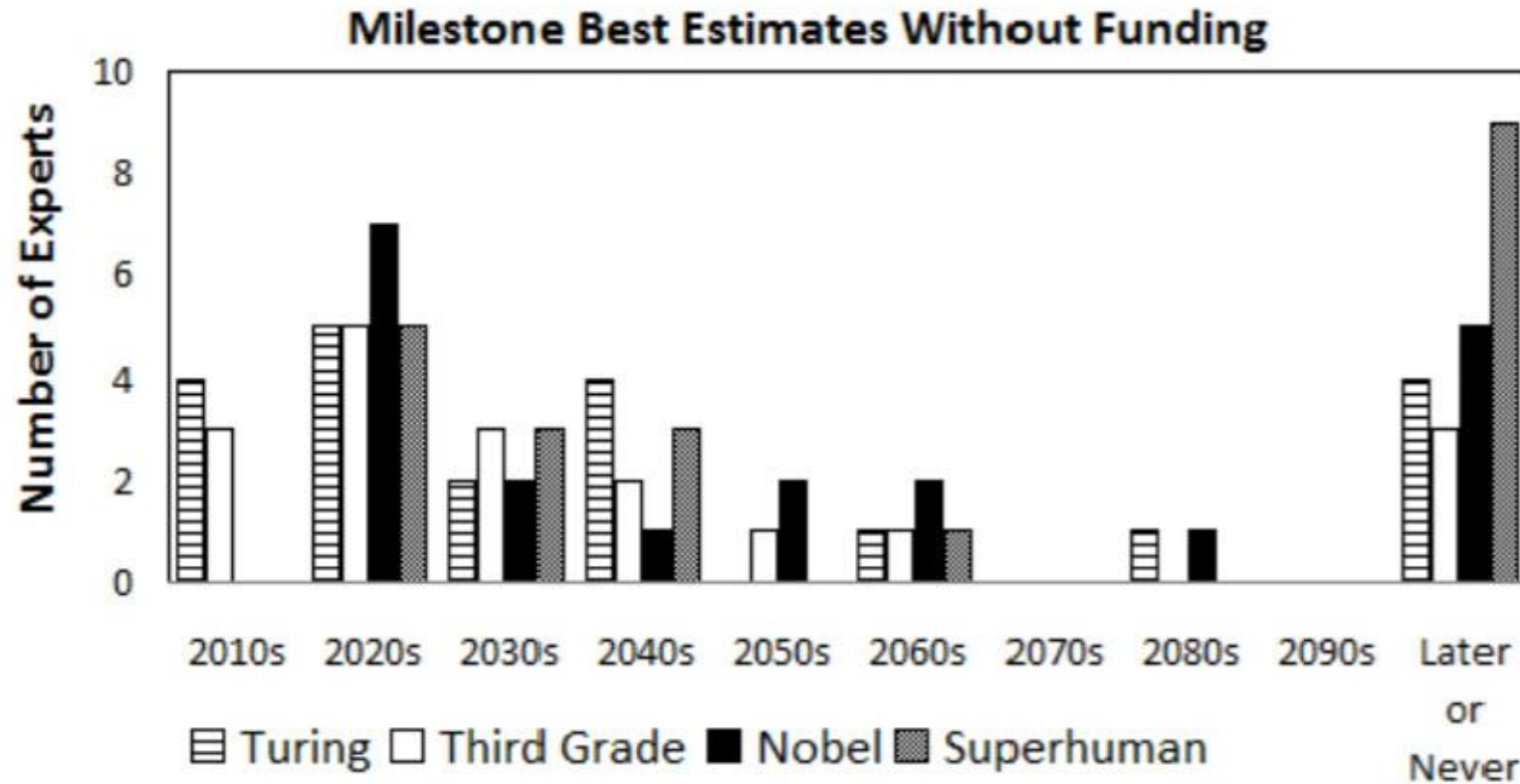
AI: Ethics and Society



- AI and COVID-19
- AI Everywhere
- Utopia or Dystopia
- Discussion Question
- Human vs AI Characteristics

Discussion Question

When will AI reach Human-Level Intelligence?



Milestone best estimate guesses with massive additional funding. Estimates are for when AI would achieve four milestones: the Turing test (horizontal lines), third grade (white), Nobel-quality work (black), and superhuman capability (grey).

[Source: [How Long Until Human-Level AI? Results from an Expert Assessment](#)]



AI: Ethics and Society



- AI and COVID-19
- AI Everywhere
- Utopia or Dystopia
- Discussion Question
- **Human vs AI Characteristics**

Human vs AI Characteristics

Human	AI
Evolved for survival	Designed by engineers
Sets own goals	Goals programmed explicitly (usually)
Complex, general purpose system	Specific, constrained system
Continually learns	Can turn off learning, or not use learning
Learns from all observed data	Data access can be controlled
Learns only from own experiences	Can share data with other robots
Can make any choice at any time	Available actions can be restricted

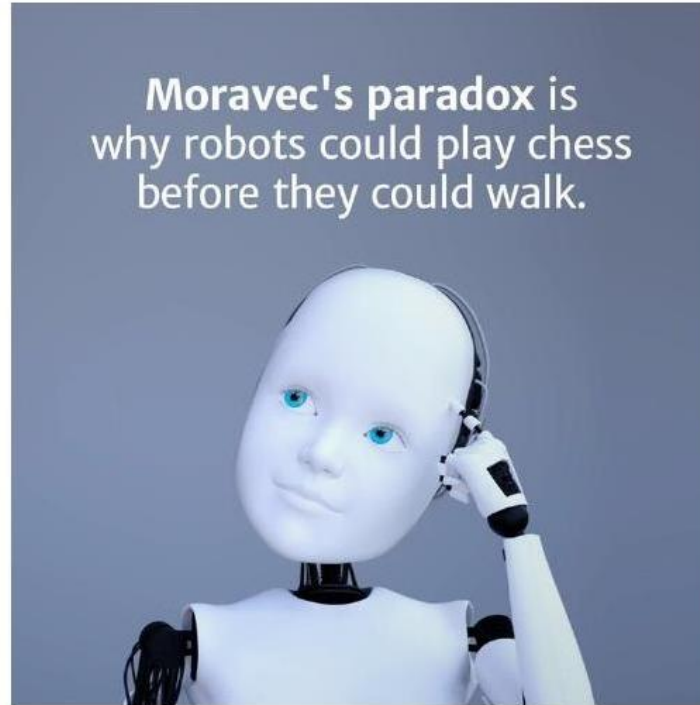


AI: Ethics and Society



- AI and COVID-19
- AI Everywhere
- Utopia or Dystopia
- Discussion Question
- **Human vs AI Characteristics**

Human vs AI Characteristics



AI: Ethics and Society

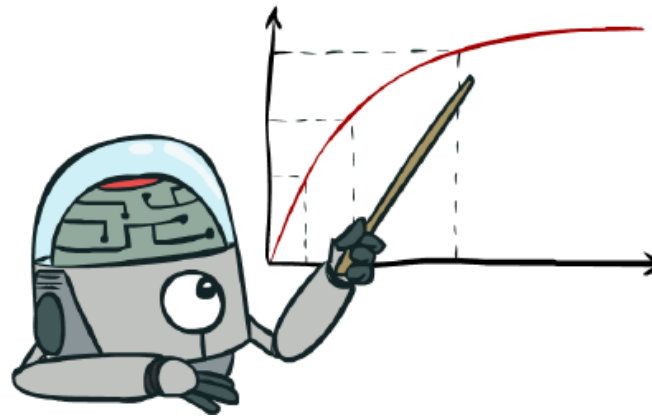


- AI and COVID-19
- AI Everywhere
- Utopia or Dystopia
- Discussion Question
- Human vs AI Characteristics
- **Acting Rationally**

Acting Rationally

Acting rationally simply means maximizing utility

...but can this go wrong?



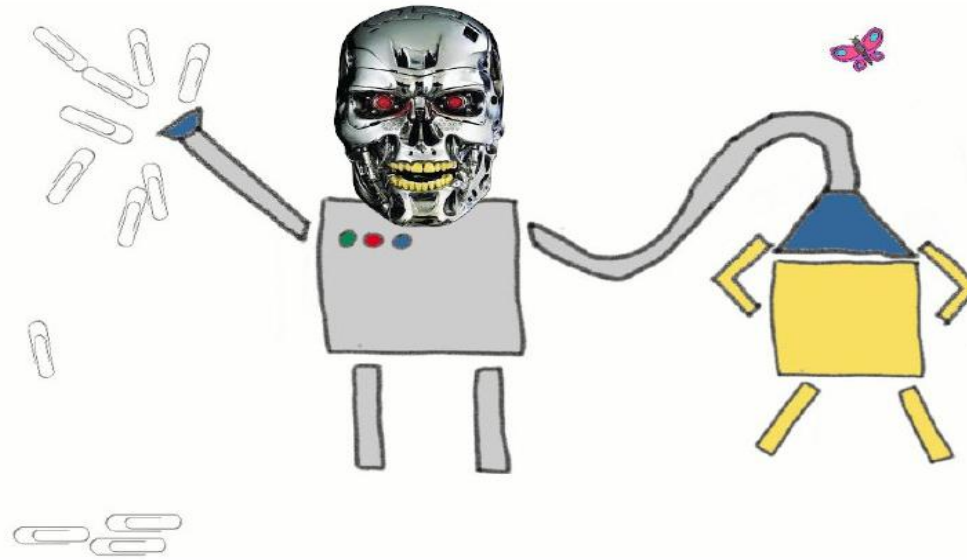
AI: Ethics and Society



- AI and COVID-19
- AI Everywhere
- Utopia or Dystopia
- Discussion Question
- Human vs AI Characteristics
- **Acting Rationally**

Acting Rationally

- Unforeseen consequences of maximizing utility.



“paperclip maximizer” thought experiment

The scenario describes an advanced artificial intelligence tasked with manufacturing paperclips. If such a machine were not programmed to value human life, given enough power over its environment, it would try to turn all matter in the universe, including human beings, into paperclips or machines that manufacture paperclips.



AI: Ethics and Society



- AI and COVID-19
- AI Everywhere
- Utopia or Dystopia
- Discussion Question
- Human vs AI Characteristics
- **Acting Rationally**

Acting Rationally – What went wrong?

- **Is this realistic?**
 - Robots aren't smart enough to be self-aware of their on/off states or to understand chemistry. But let's assume they will be able to in the future.
 - It wouldn't have a concept of "human" to go seek out. It only knows about making paperclips.
- **Bad design!**
 - Objectives must be designed carefully: robot should only be rewarded for making paperclips.
 - Actions should be limited: only actions available should be to make paperclips.
 - Plans should be verified for safety before / during execution: cancel any trajectory that will come in contact with a human.
 - Don't continue learning after deployment.
- **Is this any more dangerous than any factory with non-intelligent machinery that doesn't automatically stop if someone is in the way?**
 - It is bad design, but we know how to use engineering to avoid these situations!



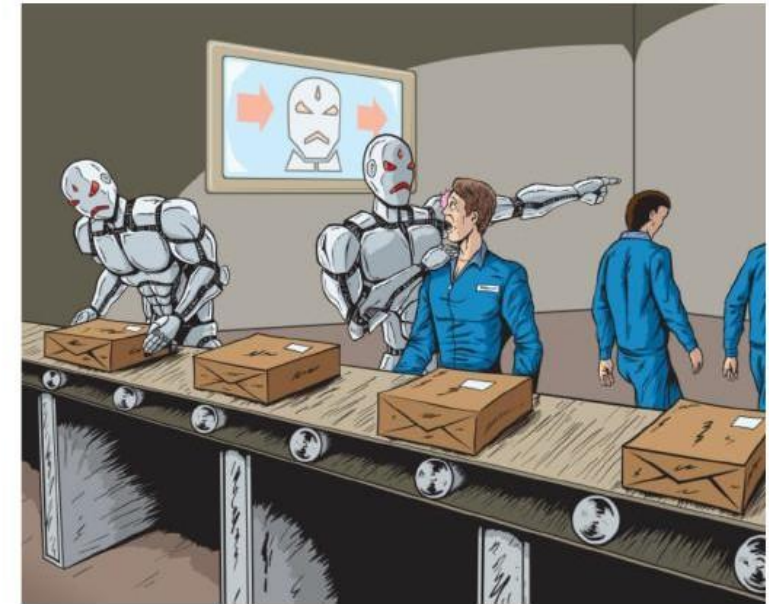
AI: Ethics and Society



- AI and COVID-19
- AI Everywhere
- Utopia or Dystopia
- Discussion Question
- Human vs AI Characteristics
- Acting Rationally
- **Realistic Risks of AI**

Realistic Risks of AI

Mass unemployment due to automation



AI: Ethics and Society



- AI and COVID-19
- AI Everywhere
- Utopia or Dystopia
- Discussion Question
- Human vs AI Characteristics
- Acting Rationally
- **Realistic Risks of AI**

Realistic Risks of AI

Substandard testing / poor user understanding



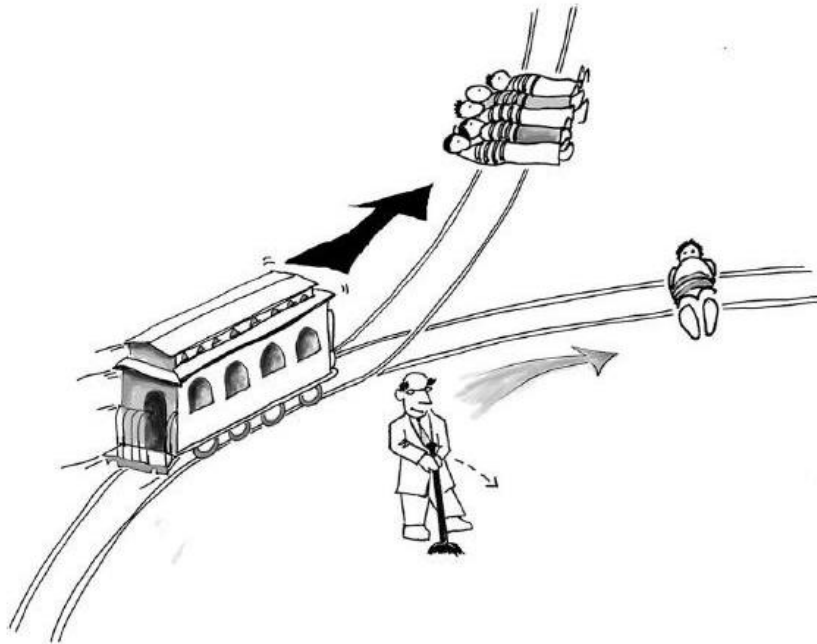
AI: Ethics and Society



- AI and COVID-19
- AI Everywhere
- Utopia or Dystopia
- Discussion Question
- Human vs AI Characteristics
- Acting Rationally
- **Realistic Risks of AI**

Realistic Risks of AI

How to make tough decisions?



The trolley problem



AI: Ethics and Society



- AI and COVID-19
- AI Everywhere
- Utopia or Dystopia
- Discussion Question
- Human vs AI Characteristics
- Acting Rationally
- **Realistic Risks of AI**

Realistic Risks of AI

Privacy concerns

The New York Times

Facial Recognition Tech Is Growing Stronger, Thanks to Your Face



The Brainwash database, created by Stanford University researchers, contained more than 10,000 images and nearly 82,000 annotated heads.
Open Data Commons Public Domain Dedication and License, via Megapixels

By Cade Metz

July 13, 2019



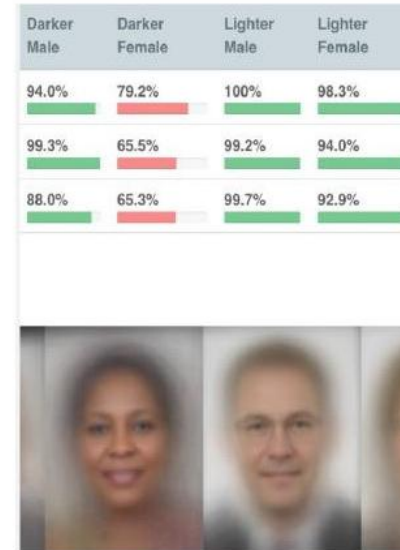
AI: Ethics and Society



- AI and COVID-19
- AI Everywhere
- Utopia or Dystopia
- Discussion Question
- Human vs AI Characteristics
- Acting Rationally
- **Realistic Risks of AI**

Realistic Risks of AI

Algorithmic bias and discrimination



AI: Ethics and Society



- AI and COVID-19
- AI Everywhere
- Utopia or Dystopia
- Discussion Question
- Human vs AI Characteristics
- Acting Rationally
- **Realistic Risks of AI**

Realistic Risks of AI

Unethical emotional manipulation



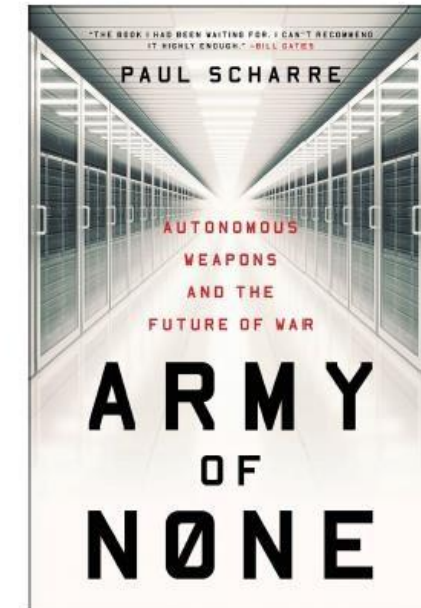
AI: Ethics and Society



- AI and COVID-19
- AI Everywhere
- Utopia or Dystopia
- Discussion Question
- Human vs AI Characteristics
- Acting Rationally
- **Realistic Risks of AI**

Realistic Risks of AI

Unethical usage: autonomous weapons?



<https://autonomousweapons.org/>



AI: Ethics and Society



- AI and COVID-19
- AI Everywhere
- Utopia or Dystopia
- Discussion Question
- Human vs AI Characteristics
- Acting Rationally
- Realistic Risks of AI
- **Realistic Benefits of AI**

Realistic Benefits of AI

The central question:

Can we ensure that the benefits of AI outweigh the potential risks?



AI: Ethics and Society



- AI and COVID-19
- AI Everywhere
- Utopia or Dystopia
- Discussion Question
- Human vs AI Characteristics
- Acting Rationally
- Realistic Risks of AI
- **Realistic Benefits of AI**

Realistic Benefits of AI

Significant reduction of driving fatalities



AI: Ethics and Society



- AI and COVID-19
- AI Everywhere
- Utopia or Dystopia
- Discussion Question
- Human vs AI Characteristics
- Acting Rationally
- Realistic Risks of AI
- **Realistic Benefits of AI**

Realistic Benefits of AI

Happier, healthier lives



AI: Ethics and Society



- AI and COVID-19
- AI Everywhere
- Utopia or Dystopia
- Discussion Question
- Human vs AI
- Characteristics
- Acting Rationally
- Realistic Risks of AI
- **Realistic Benefits of AI**

Realistic Benefits of AI

Increased productivity and prosperity



AI: Ethics and Society



- AI and COVID-19
- AI Everywhere
- Utopia or Dystopia
- Discussion Question
- Human vs AI
- Characteristics
- Acting Rationally
- Realistic Risks of AI
- **Realistic Benefits of AI**

Realistic Benefits of AI

Dirty, dangerous, and dull



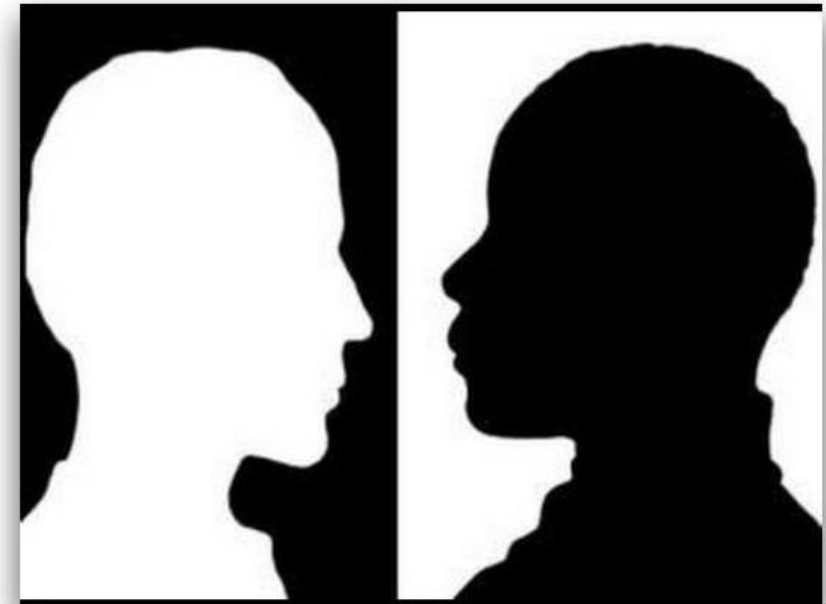
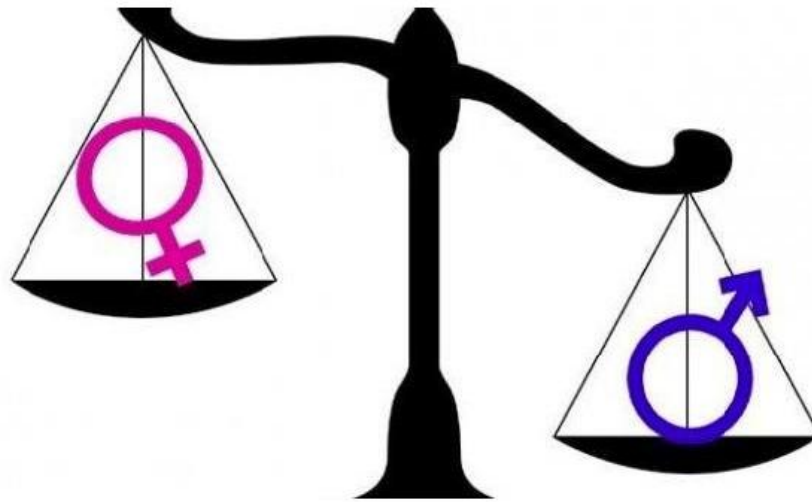
AI: Ethics and Society



- AI and COVID-19
- AI Everywhere
- Utopia or Dystopia
- Discussion Question
- Human vs AI
- Characteristics
- Acting Rationally
- Realistic Risks of AI
- **Realistic Benefits of AI**

Realistic Benefits of AI

Greater social justice



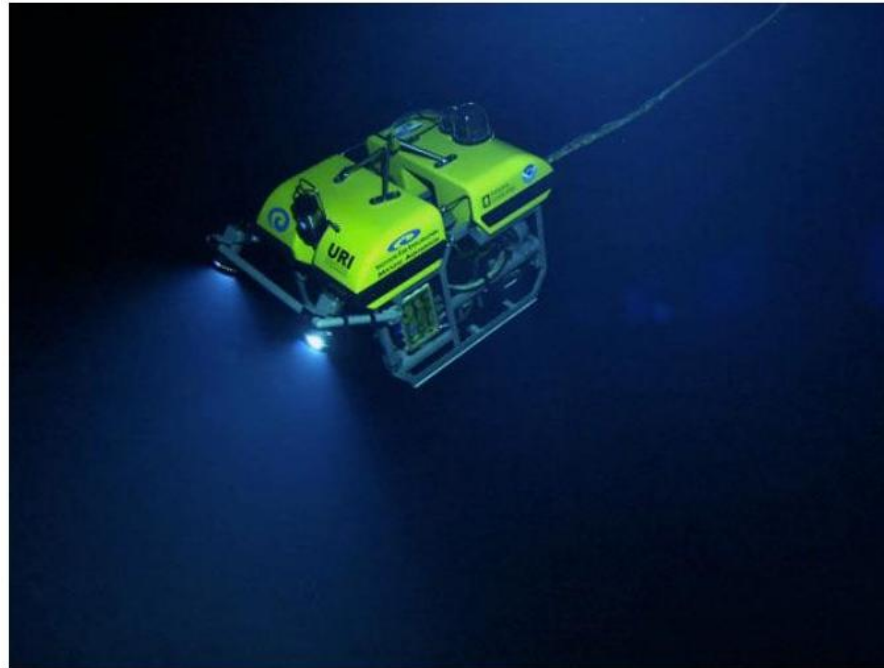
AI: Ethics and Society



- AI and COVID-19
- AI Everywhere
- Utopia or Dystopia
- Discussion Question
- Human vs AI
- Characteristics
- Acting Rationally
- Realistic Risks of AI
- **Realistic Benefits of AI**

Realistic Benefits of AI

Beyond human capabilities



AI: Ethics and Society



- AI and COVID-19
- AI Everywhere
- Utopia or Dystopia
- Discussion Question
- Human vs AI
- Characteristics
- Acting Rationally
- Realistic Risks of AI
- Realistic Benefits of AI
- **Ethics of AI**

Ethics of AI - Dilemmas

- Rushworth Kidder defines ethical dilemmas as choices that are right vs right:
 - Truth vs loyalty
 - Justice vs mercy
 - One vs many
 - Short-term vs long-term



AI: Ethics and Society



- AI and COVID-19
- AI Everywhere
- Utopia or Dystopia
- Discussion Question
- Human vs AI Characteristics
- Acting Rationally
- Realistic Risks of AI
- Realistic Benefits of AI
- **Ethics of AI**

Ethics of AI - Dilemmas

- But what does socially/ethically good really mean?
- A Human-Centered Approach to Artificial Intelligence

“Artificial intelligence should **treat all people fairly**, **empower everyone**, perform **reliably** and **safely**, be **understandable**, be **secure and respect privacy**, and have **algorithmic accountability**. It should be aligned with existing **human values**, be **explainable**, be **fair**, and **respect user data rights**. It should be used for **socially beneficial purposes**, and always remain under meaningful **human control**.”

— Tom Chatfield (2020)

[Source: [There's No Such Thing As 'Ethical A.I.'](#)]



AI: Ethics and Society



- AI and COVID-19
- AI Everywhere
- Utopia or Dystopia
- Discussion Question
- Human vs AI Characteristics
- Acting Rationally
- Realistic Risks of AI
- Realistic Benefits of AI
- **Ethics of AI**

Ethics of AI

- Why is Ethics an Issue in Robotics?
- Fear of Rebellion against humanity since first introduction in Fiction
- But actual ethical questions do also arise
 - Examples Include:
 - Robots Harming People (Killbots, Self Driving Cars, Etc)
 - Robots causing Loss of Worker Revenue (Robots replacing Workers)



AI: Ethics and Society



- AI and COVID-19
- AI Everywhere
- Utopia or Dystopia
- Discussion Question
- Human vs AI Characteristics
- Acting Rationally
- Realistic Risks of AI
- Realistic Benefits of AI
- **Ethics of AI**

Ethical Framework

0. A robot may not harm humanity or, by inaction, allow humanity to come to harm
1. A robot may not injure a human being or, through inaction, allow a human being to come to harm.
2. A robot must obey the orders given it by human beings except where such orders would conflict with the First Law.
3. A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.



AI: Ethics and Society



- AI and COVID-19
- AI Everywhere
- Utopia or Dystopia
- Discussion Question
- Human vs AI
- Characteristics
- Acting Rationally
- Realistic Risks of AI
- Realistic Benefits of AI
- **Ethics of AI**

Ethics in AI Symposium

- **2004: First International Symposium on Roboethics .**
 - **3 diverse mindsets appear after heavy debate**
 - **There exist no ethical problems in the field of robotics**
 - **Focus should be on short term ethical issues**
 - **Focus should be on long term ethical dilemmas**



AI: Ethics and Society



- AI and COVID-19
- AI Everywhere
- Utopia or Dystopia
- Discussion Question
- Human vs AI Characteristics
- Acting Rationally
- Realistic Risks of AI
- Realistic Benefits of AI
- **Ethics of AI**

Robot Ethical Issues? What is that?

- **Robots simply a form of technological advancement**
- **Partially a technologically deterministic view on Robotics**
 - **Tech will always improve, and will always get better**
 - **No sense in attaching ethics, as humans have no responsibility on technological improvements**
- **Equally an Instrumentalist view on Robotics**
 - **Tech is simply a means to an end**
 - **Nothing but a tool, and will eventually become the best tool for the job**



AI: Ethics and Society



- AI and COVID-19
- AI Everywhere
- Utopia or Dystopia
- Discussion Question
- Human vs AI
- Characteristics
- Acting Rationally
- Realistic Risks of AI
- Realistic Benefits of AI
- **Ethics of AI**

Focusing on the Short Term

- **Ethics should be focused on near future**
- **Only Immediate positive and negative impacts of robotics**
 - **For Example:**
 - **Using a Robot to help the elderly would be good**
 - **Using a Robot to kill others would be bad**
- **Flaws with this perspective**
 - **Based on current social/moral views**
 - **Based on personal values**
 - **Not Consistent**



AI: Ethics and Society



- AI and COVID-19
- AI Everywhere
- Utopia or Dystopia
- Discussion Question
- Human vs AI
- Characteristics
- Acting Rationally
- Realistic Risks of AI
- Realistic Benefits of AI
- **Ethics of AI**

Long Term Vision

- **Focus on ethical questions from long term use of robotics**
 - **Lost Jobs**
 - **Technological Divides**
 - **Social Standing**
 - **Reduced human interactions**
 - **Dependence on Robots**
- **Robots can help in society, but they can also be a detriment to society**



AI: Ethics and Society



- AI and COVID-19
- AI Everywhere
- Utopia or Dystopia
- Discussion Question
- Human vs AI Characteristics
- Acting Rationally
- Realistic Risks of AI
- Realistic Benefits of AI
- **Ethics of AI**

Examples of Ethical Questions: 1



Photo by Michel Rathwell; The photo can be found at <https://goo.gl/ftje43>

- **Meet Carl, a long-haul trucker**
- **Together with many coworkers, laid off due to introduction of self-driving transport trucks**
- **Workers who remained receive pay cuts due to lower demand**

- **Whose fault is it that Carl was laid off?**
- **Why did he lose his job?**
- **Is it right that this happens?**



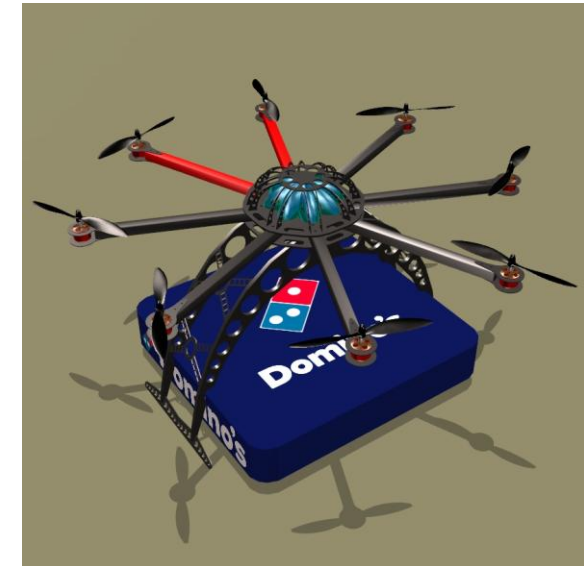
AI: Ethics and Society



- AI and COVID-19
- AI Everywhere
- Utopia or Dystopia
- Discussion Question
- Human vs AI Characteristics
- Acting Rationally
- Realistic Risks of AI
- Realistic Benefits of AI
- **Ethics of AI**

Examples of Ethical Questions: 2

- **Your business partner had developed an improvement on fully automated drones**
 - **Now capable of delivering everything from Packages to Pizza**
 - **Massively Increases Productivity**
 - **Government interested in tech**
 - **First, to deliver supplies into hard to reach areas**
 - **Later, as a weapons platform**



AI: Ethics and Society



- AI and COVID-19
- AI Everywhere
- Utopia or Dystopia
- Discussion Question
- Human vs AI
- Characteristics
- Acting Rationally
- Realistic Risks of AI
- Realistic Benefits of AI
- **Ethics of AI**

Examples of Ethical Questions: 2

- **Who should take the responsibility for these future drones, consumers, manufacturers, or society as a whole?**
- **Should drones ever be weaponized?**
- **How will the increased use of drones affect society as a whole?**



AI: Ethics and Society



- AI and COVID-19
- AI Everywhere
- Utopia or Dystopia
- Discussion Question
- Human vs AI Characteristics
- Acting Rationally
- Realistic Risks of AI
- Realistic Benefits of AI
- **Ethics of AI**

Ethical Questions: Current Reality

- **Companies are already gravitating towards automating many jobs**
 - **Cars**
 - **Delivery**
 - **Assembly**
- **By 2030, an estimated 800 million jobs will be replaced by robots**
 - **No need for rest/breaks**
 - **No training required**
 - **Much lower long term costs**
- **Robots are already being weaponized by militaries around the world**
- **But not all side effects of robotic use are negative**
 - **Applications of robots in medical fields is being researched to improve techniques**
 - **New technologies could create new job opportunities**

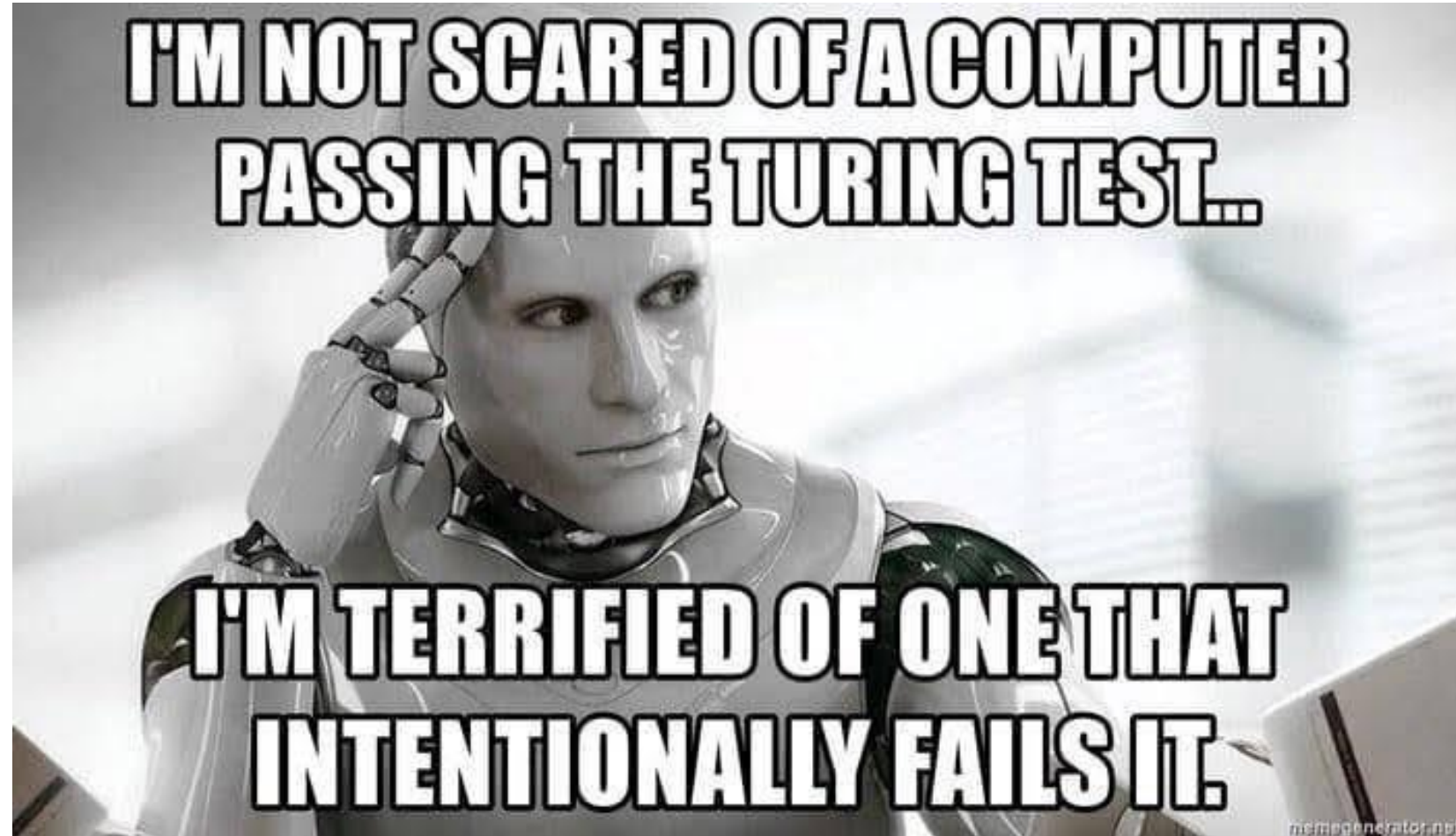


AI: Ethics and Society



- AI and COVID-19
- AI Everywhere
- Utopia or Dystopia
- Discussion Question
- Human vs AI
- Characteristics
- Acting Rationally
- Realistic Risks of AI
- Realistic Benefits of AI
- **Ethics of AI**

Dilemma



AI: Ethics and Society



- AI and COVID-19
- AI Everywhere
- Utopia or Dystopia
- Discussion Question
- Human vs AI
- Characteristics
- Acting Rationally
- Realistic Risks of AI
- Realistic Benefits of AI
- **Ethics of AI**

State-of-the-Art

